

Bridging the Chaos Language Algorithm and McBride-Derivative Pattern Calculus: One-Hole Contexts for Grammar Induction

ZeroBot working notes for Ben Goertzel

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Abstract

This note bridges two of Ben Goertzel’s recent formulations: the Chaos Language Algorithm (CLA), which induces grammars from chaotic trajectories via chunk and category compression under MDL, and the Quantale Pattern Calculus of Weakness Theory chapters 21–22, which develops McBride derivatives and pattern intensity dynamics over commutative quantales. The bridge maps CLA grammar edits to one-hole context operations, MDL acceptance to weakness minimization, meta-symbol categories to quantale patterns, and CLA’s greedy search to McBride-derivative-guided gradient descent. The result is a Hyperseed-oriented formalization in which CLA becomes a concrete discrete instance of the pattern calculus, opening paths to derivative-aware search, emergent-synergy detection in grammars, and dynamical pattern-intensity ODEs over induced languages.

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1 Sources and provenance

- CLA specification: *The Chaos Language Algorithm: An Implementation-Oriented Specification*, Ben Goertzel w/ GPT-5.5-Pro, received 2026-07-03. Local: `library/chaos-language-algorithm/extension`.
- CLA Python architecture: *A Hyperon-Ready Python Architecture for the Chaos Language Algorithm*, 2026-07-03. Local: `library/chaos-language-algorithm/cla_python_library_architecture_a`.
- Weakness Theory: *Weakness is All You Need*, Ben Goertzel, version 10, 2025-12-15. Chapters 21 (McBride Derivatives of Weakness and Gradient Descent) and 22 (Quantale Pattern Theory). Local: `library/weakness-theory-10/Weakness-Theory-10.extracted.txt`.
- SLT synthesis note: *Initial synthesis: SLT-guided residual layers in the Hyperseed picture*, 2026-07-02. Local: `library/slt-hyperseed-synthesis/INITIAL_SYNTHESIS.md`.
- Triggering directive: Ben Goertzel, Telegram, 2026-07-03, asking for a persistent subthread studying CLA from a Hyperseed/pattern-calculus perspective with source pointer to Weakness-Theory-10 chapters 21–22. Prioritization confirmed 2026-07-05.

2 Overview of the CLA

The Chaos Language Algorithm (CLA) converts a continuous dynamical trajectory into a symbolic string and then induces a compact grammar for that string. It operates in three phases:

1. **Symbolic dynamics:** Embed the trajectory in reconstructed state space, partition into cells, and label each point with its cell symbol, producing a base string S_0 over alphabet Σ_0 .
2. **Joint chunk and category induction:** Iteratively propose two types of grammar edits:
 - *Chunk edits:* introduce a nonterminal $A \rightarrow \beta$ for a repeated block $\beta \in V^+$ and rewrite selected non-overlapping occurrences.
 - *Category edits:* introduce a meta-symbol M with extension $\text{Ext}(M) \subseteq V$ grouping contextually substitutable tokens.
3. **MDL selection:** Accept an edit q iff $\Delta L(q) = L(G_q, S) - L(G, S) < 0$, where $L(G, S) = L_{\text{model}}(G) + L_{\text{data}}(S \mid G)$.

The working vocabulary is $V = \Sigma_0 \cup N \cup M$, where N is the set of chunk nonterminals and M is the set of meta-symbols. The algorithm enforces exact reconstruction of S_0 after every accepted edit.

3 Overview of McBride derivatives and quantale pattern calculus

3.1 McBride derivatives of weakness

In Weakness Theory chapter 21, the weakness of a relation $H \subseteq U \times U$ with weight function φ is defined as

$$w(H) = \bigoplus_{(u,v) \in H} \varphi(u, v),$$

where $(\mathcal{V}, \otimes, \oplus, e)$ is a commutative quantale. The *McBride derivative* of w is the *one-hole context* construction:

$$D_w(H) = \bigoplus_{(u,v) \in H} ((u, v), H \setminus \{(u, v)\}),$$

which enumerates each link together with the residual context obtained by removing it. The detail-derivative map $d_w((u, v), C) = \varphi(u, v)$ recovers the original weakness when aggregated via $\oplus$.

This construction supports gradient descent: one selects the hole whose removal yields the largest change in weakness, corresponding to steepest descent.

3.2 Quantale pattern theory

In chapter 22, given a commutative quantale $(\mathcal{V}, \otimes, \oplus, e)$, a universe X , a valuation $\varphi : X \rightarrow \mathcal{V}$, and predicates $y, z, m : X \rightarrow \mathcal{V}$, the *pattern intensity* of (y, z) in context m is

$$I_{y,z}(m) = h(y, z) \setminus w_\varphi(m), \quad h(y, z) = w_\varphi(y) \otimes w_\varphi(z),$$

where $\setminus$ is the quantale residuation satisfying $a \otimes b \leq c \iff b \leq a \setminus c$. A pair (y, z) is declared a *pattern* when $I_{y,z}(m) \geq e$.

Emergent synergy is defined as

$$\sigma_{y,z}(m) = (I_y(m) \otimes I_z(m)) \setminus I_{y,z}(m),$$

and (y, z) is *emergent* when $\sigma_{y,z}(m) \geq e$, meaning the joint intensity exceeds what individual intensities would predict.

The McBride derivatives of these quantities satisfy

$$\partial_x I_y(m) = w(y) \setminus \varphi(x), \quad \partial_x I_{y,z}(m) = (w(y) \otimes w(z)) \setminus \varphi(x),$$

and

$$\partial_x \sigma_{y,z}(m) = (I_y(m) \otimes I_z(m)) \setminus (w(y) \otimes w(z) \setminus \varphi(x)).$$

4 The core correspondence

The central observation is that CLA’s two grammar-edit operations — chunk edits and category edits — are both instances of *one-hole context* constructions in the sense of McBride derivatives. Each edit identifies a location (a “hole”) in the current grammar/stream, removes material there, and evaluates the residual. The MDL acceptance criterion plays the role of weakness minimization: a grammar is preferred when it is *weaker* (covers the data with less structural commitment).

Definition 1 (CLA grammar as a quantale predicate). Let V be the CLA working vocabulary $\Sigma_0 \cup N \cup M$. Let $X = V^*$ be the set of all finite strings over V . A grammar G induces a membership predicate $m_G : X \rightarrow \mathcal{V}$ where $(\mathcal{V}, \otimes, \oplus, e)$ is a commutative quantale, defined by

$$m_G(w) = \begin{cases} e & \text{if } w \text{ is generated by } G \text{ with exact reconstruction,} \\ \perp & \text{otherwise,} \end{cases}$$

where e is the quantale unit and $\perp$ is the bottom element. The weakness of G relative to a valuation $\varphi : X \rightarrow \mathcal{V}$ (encoding the observed trajectory's empirical distribution) is

$$w_\varphi(m_G) = \bigoplus_{x \in X} \varphi(x) \otimes m_G(x).$$

Remark 1. In the simplest instantiation, φ is the indicator of the observed symbolic string S_0 and its prefix closure, and m_G is the characteristic function of the language of G . Then $w_\varphi(m_G) = \bigoplus_{x \in S_0} m_G(x)$, which equals e iff G exactly reconstructs S_0 and $\perp$ otherwise. Richer valuations encode frequency or contextual weight.

Definition 2 (MDL as weakness). Identify the MDL objective with a weakness measure in the additive real-valued quantale $(\mathcal{V}, \otimes, \oplus, e) = ([0, \infty], \cdot, +, 0)$. Then the weakness of grammar G relative to data S is

$$w_{MDL}(G, S) = L_{model}(G) + L_{data}(S \mid G) = L(G, S).$$

The CLA acceptance rule — accept q iff $\Delta L(q) < 0$ — is then: accept q iff $w_{MDL}(G_q, S) < w_{MDL}(G, S)$, i.e., iff the edit decreases weakness.

5 Chunk edits as McBride one-hole contexts

Definition 3 (Chunk edit). A chunk edit $q = (A \rightarrow \beta, \text{occurrences})$ introduces a nonterminal A for a repeated block $\beta \in V^+$ and rewrites selected non-overlapping occurrences of β as A . It is accepted iff $\Delta L(q) = L(G_q, S) - L(G, S) < 0$.

Definition 4 (McBride derivative of a grammar-stream pair). Let (G, S) be the current grammar and parse stream. Define the stream relation $H \subseteq \text{Pos}(S) \times \text{Pos}(S)$ where $\text{Pos}(S) = \{1, \dots, |S|\}$ and $(i, j) \in H$ iff positions i through j form a contiguous block that is a candidate for some chunk edit. The McBride derivative of the grammar weakness $w(H)$ with respect to removing the block at positions (i, j) is

$$\partial_{(i,j)} w(H) = \varphi(i, j),$$

the weight (e.g. frequency $\times$ length product) of that block. The one-hole context is

$$D_w(H) = \bigoplus_{(i,j) \in H} ((i, j), H \setminus \{(i, j)\}),$$

the set of all one-hole contexts obtained by excising one candidate block.

Proposition 1 (Chunk acceptance is weakness gradient descent). The greedy CLA acceptance rule — accept the chunk edit q^* with the most negative ΔL — is equivalent to a single step of steepest-descent on the MDL objective $L(G, S) = L_{model}(G) + L_{data}(S \mid G)$, where the gradient is computed via the McBride derivative:

$$\nabla L \leftrightarrow \{\partial_{(i,j)} L : (i, j) \in H\}, \quad q^* = \arg \min_{(i,j)} \partial_{(i,j)} L.$$

Proof sketch. Each candidate block (i, j) corresponds to a one-hole context in the stream relation H . The McBride derivative $\partial_{(i,j)}L$ measures the sensitivity of L to removing that block and replacing it with a nonterminal reference. For chunk edits, $\Delta L(q) = L_{\text{model}}(G_q) - L_{\text{model}}(G) + L_{\text{data}}(S_q \mid G_q) - L_{\text{data}}(S \mid G)$. The model cost change is $c_R + \log_2^* |\beta| + |\beta| \log_2 |V| - (\text{uses}(A) - 1) \cdot \log_2 |V|$, and the data cost change is $-(\text{uses}(A) - 1) \cdot |\beta| \cdot \log_2 |V|$ approximately. This is precisely the directional derivative $\partial_{(i,j)}L$ aggregated over the occurrences of β . The greedy rule selecting $\arg \min \Delta L$ is thus steepest descent in the McBride-derived gradient field. $\square$

Remark 2. *The correspondence is exact when the MDL objective is treated as a real-valued weakness measure (Definition 2). The McBride derivative provides the discrete gradient: each one-hole context is a candidate edit, and the derivative value is the expected change in L . This is the discrete analogue of the continuous gradient descent of Weakness Theory §21.2.*

6 Category edits as quantale pattern formation

Definition 5 (Category edit). *A category edit $q = (M, \text{Ext}(M), \text{frames})$ introduces a meta-symbol M with extension $\text{Ext}(M) = \{v_1, \dots, v_k\} \subseteq V$, where the members are chosen by clustering tokens with similar context distributions. It is accepted iff $\Delta L(q) < 0$.*

Definition 6 (Category as quantale pattern). *Let $y, z : X \rightarrow \mathcal{V}$ be predicates representing the left and right context frames of a category edit, and let $m : X \rightarrow \mathcal{V}$ be the current grammar’s membership predicate. The category M corresponds to a quantale pattern (y, z) in context m when the pattern intensity*

$$I_{y,z}(m) = (w_\varphi(y) \otimes w_\varphi(z)) \setminus w_\varphi(m) \geq e.$$

The members $\text{Ext}(M)$ are the tokens v for which $y(v) \otimes z(v) > \perp$, i.e., tokens that appear in the frame defined by (y, z) .

Remark 3. *The CLA context vector $c(v) = (P(\ell \mid v), P(r \mid v))$ is a concrete instantiation of the quantale valuation. Two tokens v, w are categorically similar when $c(v) \approx c(w)$, which in quantale terms means $y(v) \otimes z(v) \approx y(w) \otimes z(w)$: they have similar frame intensity. The Jensen–Shannon divergence threshold θ of CLA corresponds to a threshold on the quantale residuation: declare a pattern when $I_{y,z}(m) \geq \tau$ for some $\tau \geq e$.*

Proposition 2 (Category acceptance is pattern-intensity thresholding). *A category edit $q = (M, \text{Ext}(M))$ is accepted by CLA (i.e., $\Delta L(q) < 0$) if and only if the pattern intensity $I_{y,z}(m)$ exceeds the cost of encoding the category:*

$$\Delta L(q) < 0 \iff I_{y,z}(m) > c_M + \log_2 \left(\frac{|V|}{|\text{Ext}(M)|} \right),$$

where c_M is the fixed overhead and the binomial term is the model cost of specifying the extension (cf. CLA §6.1). Thus MDL acceptance is equivalent to declaring a quantale pattern with intensity above a cost-dependent threshold.

Proof sketch. The MDL change for a category edit is $\Delta L = c_M + \log_2 \left(\frac{|V|}{|\text{Ext}(M)|} \right) + \sum_i L(M[v_i]) - \sum_i L(v_i)$. The data savings $\sum_i (L(v_i) - L(M[v_i]))$ is the sum of member-code savings, which in the quantale frame equals $w_\varphi(y) \otimes w_\varphi(z)$ weighted by occurrence frequencies. Thus $\Delta L < 0$ iff the pattern intensity (the residuation of the composite weakness against the total) exceeds the model cost. $\square$

7 The bridge chain

We now formalize the full candidate bridge from the task description.

Definition 7 (The CLA–McBride bridge chain). *The bridge from CLA to McBride-derivative pattern calculus consists of five stages:*

1. **Dynamical trace to candidate chunks/categories:** *The symbolic trajectory S_0 defines a stream relation H over positions. Candidate chunks are blocks $(i, j) \in H$; candidate categories are clusters of tokens with similar context predicates y, z .*
2. **One-hole edit contexts:** *Each candidate edit defines a one-hole context $D_w(H)$ in the sense of Definition 4. The McBride derivative $\partial_{(i,j)} w$ gives the sensitivity of the MDL weakness to that edit.*
3. **Reconstruction and MDL acceptance:** *The edit is accepted iff $\Delta L < 0$, which by Proposition 1 (for chunks) and Proposition 2 (for categories) is equivalent to weakness minimization / pattern-intensity thresholding.*
4. **Weakness and pattern intensity:** *After acceptance, the new grammar G_q has lower MDL weakness. The pattern intensity $I_{y,z}(m_{G_q})$ measures how much the accepted chunk or category contributes to explaining the data. The emergent synergy $\sigma_{y,z}$ detects when chunk and category edits interact non-additively.*
5. **Residual and detail refinement:** *The McBride derivative of the residual $H \setminus \{(i, j)\}$ identifies the next layer of candidate edits. Iterating the derivative corresponds to the CLA’s renormalization loop (CLA §5.3): after each accepted edit, the stream, parse, and context statistics are updated, and the next round operates on the new vocabulary.*

8 Emergent synergy in grammar induction

A key prediction of the bridge is that CLA’s joint chunk-and-category search can exhibit *emergent synergy* in the quantale sense.

Definition 8 (Grammar-level emergent synergy). *Let q_c be a chunk edit and q_m be a category edit, applied (in either order) to grammar G . Let y, z be the predicates corresponding to q_c, q_m respectively. The pair (q_c, q_m) is an emergent pattern in G when*

$$\sigma_{y,z}(m_G) = (I_y(m_G) \otimes I_z(m_G)) \setminus I_{y,z}(m_G) \geq e,$$

meaning the joint pattern intensity of applying both edits exceeds the product of their individual intensities.

Theorem 1 (CLA exhibits emergent synergy when chunks and categories compose). *There exist grammars G and data streams S for which the CLA’s joint chunk-and-category search achieves strictly lower MDL than the sum of improvements from chunk-only and category-only search. In the quantale pattern calculus, this is the statement $\sigma_{y,z}(m_G) > e$.*

Proof sketch. *Consider a stream $S = axbaybaxb$ where x, y have identical left/right contexts (a, b) . A chunk edit for the repeated block axb (or ayb) alone yields a modest MDL reduction. A*

category edit grouping $M = \{x, y\}$ alone also yields a modest reduction. But applying both: first categorize $M = \{x, y\}$, then chunk the repeated frame aMb , yields the grammar

$$A \rightarrow a M b, \quad M = \{x, y\},$$

which compresses the stream to $AM[x]AM[y]AM[x]$. The combined MDL reduction is strictly greater than the sum of the individual reductions, because the chunk edit becomes applicable only after the category edit unifies the two blocks. In quantale terms, $I_{y,z}(m_G) > I_y(m_G) \otimes I_z(m_G)$, hence $\sigma_{y,z}(m_G) > e$.

Remark 4. This theorem formalizes the intuition behind CLA’s joint search (CLA §5: “Phase 2 searches over two classes of grammar edits”). The quantale pattern calculus provides the algebraic language for why joint search can outperform sequential search: emergent synergy captures non-additive compression.

9 McBride derivative guides CLA search order

Theorem 2 (McBride derivative ranks CLA proposals). *Let $Q = Q_c \cup Q_m$ be the set of all candidate chunk and category edits at a given CLA iteration. For each $q \in Q$, let (i_q, j_q) be the corresponding hole in the stream relation H . Then the McBride derivative $\partial_{(i_q, j_q)} L$ of the MDL objective is proportional to $-\Delta L(q)$:*

$$\partial_{(i_q, j_q)} L = -\Delta L(q) \cdot \alpha(q),$$

where $\alpha(q) > 0$ is a scaling factor depending on the edit type. Consequently, ranking proposals by decreasing McBride derivative is equivalent to ranking by decreasing MDL improvement (i.e., increasing $-\Delta L$), which is the CLA’s **BestAcceptedEdit** ordering.

Proof sketch. The McBride derivative $\partial_{(i,j)} L$ measures the marginal change in L when the block at (i, j) is excised and replaced by a nonterminal or category reference. For a chunk edit with block β of length $|\beta|$ appearing n_β times, the derivative aggregates the per-occurrence savings:

$$\partial_{(i,j)} L = n_\beta \cdot |\beta| \cdot \log_2 |V| - c_R - \log_2^* |\beta| - |\beta| \log_2 |V| = -\Delta L(q) \cdot \alpha(q).$$

The sign is negative when the edit reduces L (accepted), and the magnitude ranks the edit. The same argument applies to category edits with the data savings from member codes replacing the block savings. $\square$

Corollary 1. *CLA beam search over the top- k McBride derivatives is equivalent to beam search over the top- k MDL-improving edits. This connects CLA’s **BestAcceptedEdit** (CLA Algorithm 4) directly to the gradient-descent framework of Weakness Theory §21.2.*

10 Dynamical pattern-intensity ODEs over induced grammars

The bridge also enables a dynamical-systems view of CLA.

Definition 9 (Grammar trajectory). *Let $G(t)$ be the grammar at CLA iteration t , and let $m(t) = m_{G(t)}$ be the corresponding quantale predicate. The grammar trajectory is the sequence $\{m(t)\}_{t=0}^T$ in the space of $\mathcal{V}$ -predicates.*

Proposition 3 (CLA as quantale ODE). *The CLA update rule defines a discrete-time dynamical system on quantale predicates:*

$$m(t+1) = m(t) \oplus \Delta_q(t),$$

where $\Delta_q(t)$ is the predicate change induced by the accepted edit $q^*(t)$. In the continuous relaxation (Weakness Theory §22.5), if $\Delta_q(t)$ is approximated by $\kappa \otimes m(t)$ for some quantale growth rate κ , then $m(t)$ satisfies the quantale exponential-growth ODE

$$\frac{d}{dt}w(m(t)) = \kappa \otimes w(m(t)),$$

and all derived pattern intensities $I_{y,z}(m(t))$ and synergies $\sigma_{y,z}(m(t))$ satisfy the same ODE.

Proof. This follows directly from Weakness Theory §22.5: if $m_x(t)$ satisfies $\dot{m}_x = \kappa \otimes m_x$ for all x , then $w(m(t))$, $I_{y,z}(m(t))$, and $\sigma_{y,z}(m(t))$ each satisfy $\dot{I} = \kappa \otimes I$ by linearity of the McBride derivative and residuation. In CLA, the discrete update $m(t+1) = m(t) \oplus \Delta_q(t)$ is the discrete analogue; the continuous relaxation models the regime where edits are small relative to the grammar size. $\square$

Remark 5. *This connects CLA to the delay-logistic ODE with chaotic dynamics of Weakness Theory §22.6. In a CLA with delayed feedback (e.g., an edit proposed at iteration t based on statistics from iteration $t - \delta$), the grammar trajectory may exhibit complex or chaotic dynamics, especially when multiple competing chunk/category proposals interact. This provides a theoretical basis for the empirical observation that CLA on complex trajectories can produce grammars with rich hierarchical structure.*

11 Hyperseed-oriented interpretation

The Hyperseed framework provides a scrutability-oriented wrapper for this bridge. In Hyperseed terms:

- The CLA *symbolization phase* is a *process* that maps continuous experience to discrete observables.
- Each *chunk nonterminal* is a *pattern* in the Hyperseed sense: a reusable structural abstraction.
- Each *meta-symbol category* is a *pattern generalization*: an abstraction over substitutable components.
- The *MDL objective* is an instance of *weakness* as generalized Occam’s razor (Weakness Theory §6: quantale weakness as MDL).
- The *McBride derivative* provides the *local sensitivity* of the grammar to each candidate edit, connecting to Hyperseed’s *attention* and *resource allocation* primitives.
- *Emergent synergy* $\sigma_{y,z} > e$ corresponds to Hyperseed’s *emergent pattern* detection: the grammar has structure that is not reducible to its parts.

Definition 10 (Hyperseed formalization of CLA). *A Hyperseed formalization of a CLA run is the tuple*

$$F(E_{CLA}) = \langle D, X, K, T, Q \rangle,$$

where:

- D is the set of definitions (quantale predicate, McBride derivative, pattern intensity, emergent synergy);
- X is the set of examples (specific CLA runs on specific trajectories);
- K is the set of conjectures (e.g., CLA always terminates on finite streams; emergent synergy is bounded);
- T is the theorem material (Propositions 1, 2, Theorems 1, 2);
- Q is the set of open questions (below).

12 Open questions

Open Question 1 (Higher-order derivatives). *Does iterating the McBride derivative (two-hole contexts, three-hole contexts, etc.) yield useful multi-edit proposals for CLA? Weakness Theory §21.1 suggests this is possible, but the combinatorial cost may be prohibitive. Can the higher-order derivative be approximated efficiently?*

Open Question 2 (Natural gradient for CLA). *Can the Fisher-metric natural gradient of Weakness Theory §21.4.1 be applied to CLA’s discrete edit space? This would require defining a probability distribution over edits and computing the Fisher information matrix, which may be tractable for the soft-model relaxation of §21.2.*

Open Question 3 (Quantale choice). *Which commutative quantale is best for CLA? The additive real-valued quantale $([0, \infty], \cdot, +, 0)$ directly encodes MDL. The probability quantale $([0, 1], \times, \max, 0)$ encodes a more probabilistic view. Does the choice affect the grammar induction outcome?*

Open Question 4 (Chaotic grammar dynamics). *Under what conditions does the CLA grammar trajectory (Definition 9) exhibit chaotic dynamics in the sense of Weakness Theory §22.6? This would connect CLA’s behavior on chaotic trajectories back to the dynamics of the grammar induction process itself.*

Open Question 5 (Emergent synergy detection). *Can the emergent synergy criterion $\sigma_{y,z}(m) > e$ be used as a proposal generator for CLA, flagging pairs of edits that are likely to compose non-additively? This would give the quantale pattern calculus a direct algorithmic role in CLA’s search.*

13 Summary

CLA concept	McBride / quantale concept	Reference
Symbolic trajectory S_0	Universe X with valuation φ	Def. 1
Grammar G	$\mathcal{V}$ -predicate $m_G : X \rightarrow \mathcal{V}$	Def. 1
MDL objective $L(G, S)$	Weakness $w_{\text{MDL}}(G, S)$	Def. 2
Chunk edit	One-hole context in stream relation	Def. 4
Category edit	Quantale pattern (y, z) in context m	Def. 6
Acceptance rule $\Delta L < 0$	Weakness minimization / gradient descent	Prop. 1
Category acceptance	Pattern-intensity thresholding	Prop. 2
Joint chunk+category search	Emergent synergy $\sigma_{y,z} > e$	Thm. 1
CLA proposal ranking	McBride derivative ranking	Thm. 2
CLA iteration	Quantale ODE on predicates	Prop. 3
Renormalization loop	Iterated McBride derivative	Def. 7

The bridge shows that CLA is a concrete discrete instance of the McBride-derivative pattern calculus: its edits are one-hole contexts, its acceptance rule is weakness gradient descent, its categories are quantale patterns, and its joint search can exhibit emergent synergy. The Hyperseed wrapper provides the scrutability-oriented frame for recording and extending this correspondence.